Distributions of Instructional Metrics

Code
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1 A General Note on Instructional Metrics

Most instructional metrics are **not** normally distributed with a nice Gaussian curve. The growth share metrics are likely the closest to normal. The others are all positively-skewed except *average retention rate*, which has a negative skew, due to it being a binomial distribution where p = 0.86.

With all of these metrics, there is no inferential interpretation being made: the values are known descriptives, rather than estimates of a parameter. Our use of the z-score was as a simple scaling mechanism to aid in creating a centroid (i.e., the instructional averages). Neither positive nor negative z-scores carry any interpretation other than *distance from the mean in standard deviations*. A negative score does not imply that a unit does something poorly, nor does a positive imply the opposite interpretation. Likewise, no normal probability interpretation is being made with these scores.

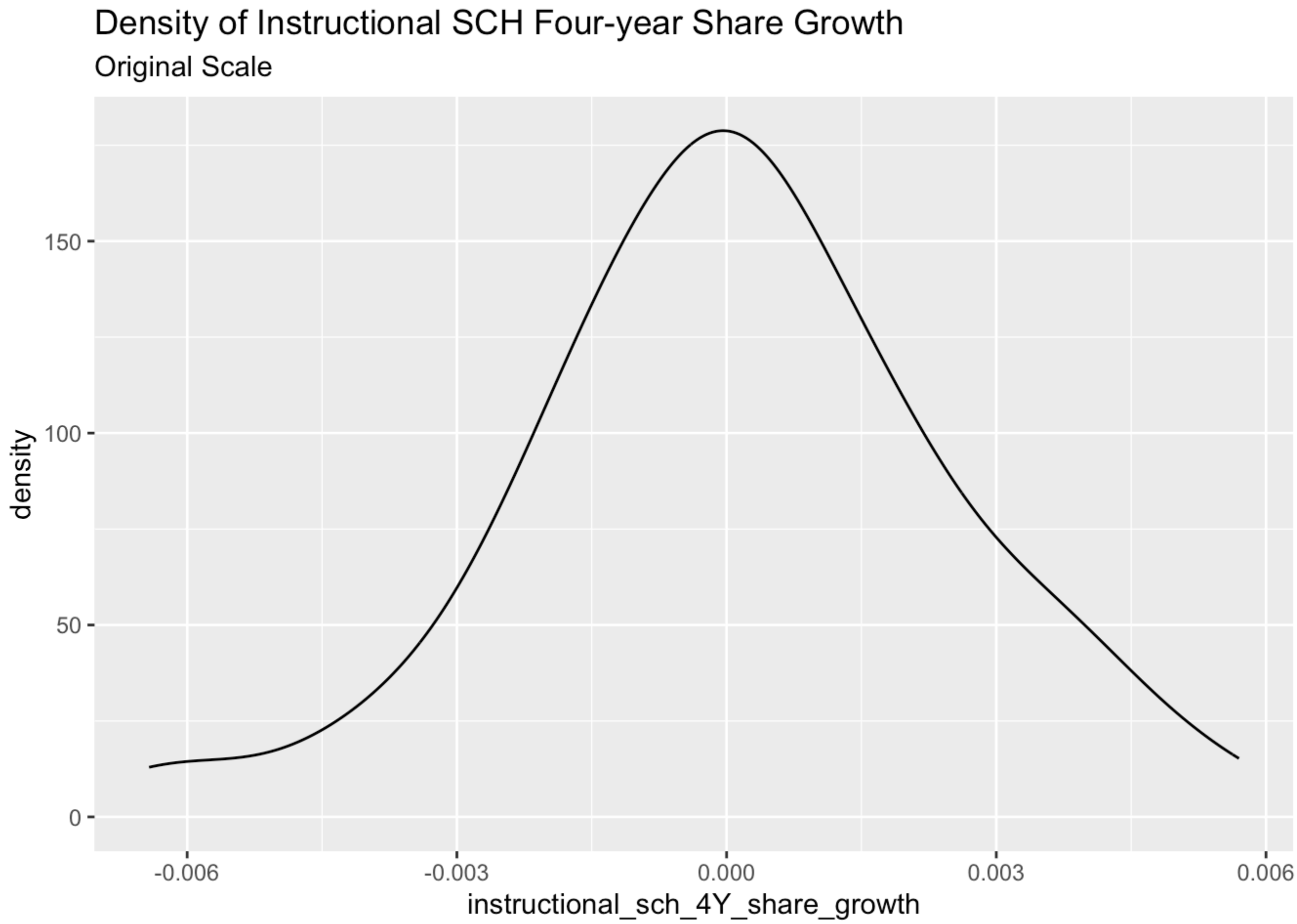
We considered transforming several of the metrics before standardization, but it was decided that intervals mattered in the interpretation. Outlying units with very large or small values should be reflected as such.

We have applied a median-based standardization approach (the mad-scores below) at the suggestion of CAS. This method expresses the unit's difference from the median (rather than mean) in median absolute deviations (MAD), scaled such that the MAD is interpretable as a pseudo standard deviation using a scaling factor estimated as sd(x) / mad(x, constant = 1). This transformation does not change value ordering and the intervals between units are proportional to the z-score.

2 Density Distributions by Variable

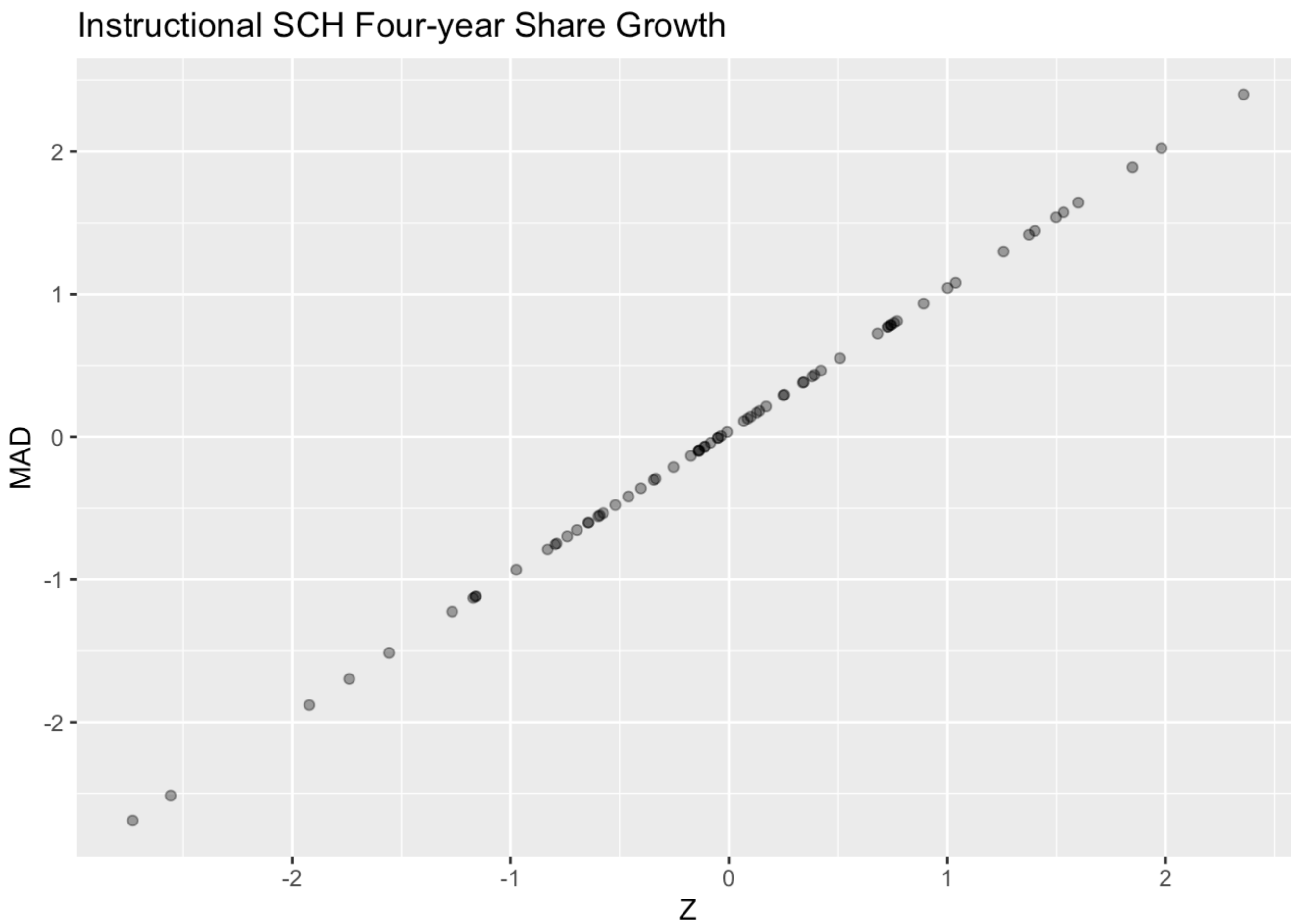
2.1 Instructional SCH Four-year Share Growth

2.1.1 Original Scale



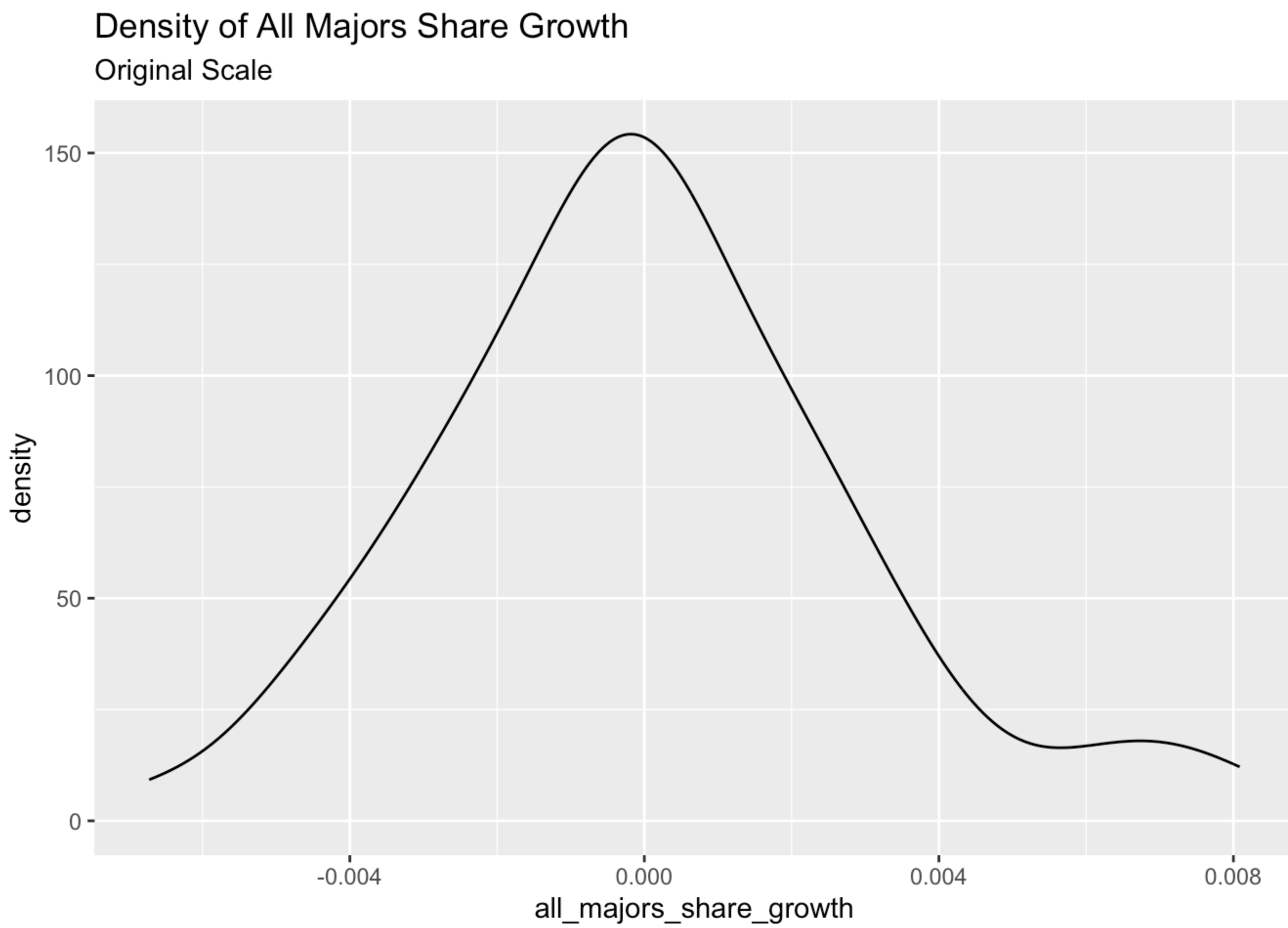
No transformed distribution shown for this metric.

2.1.2 Relationship of Z to Mad



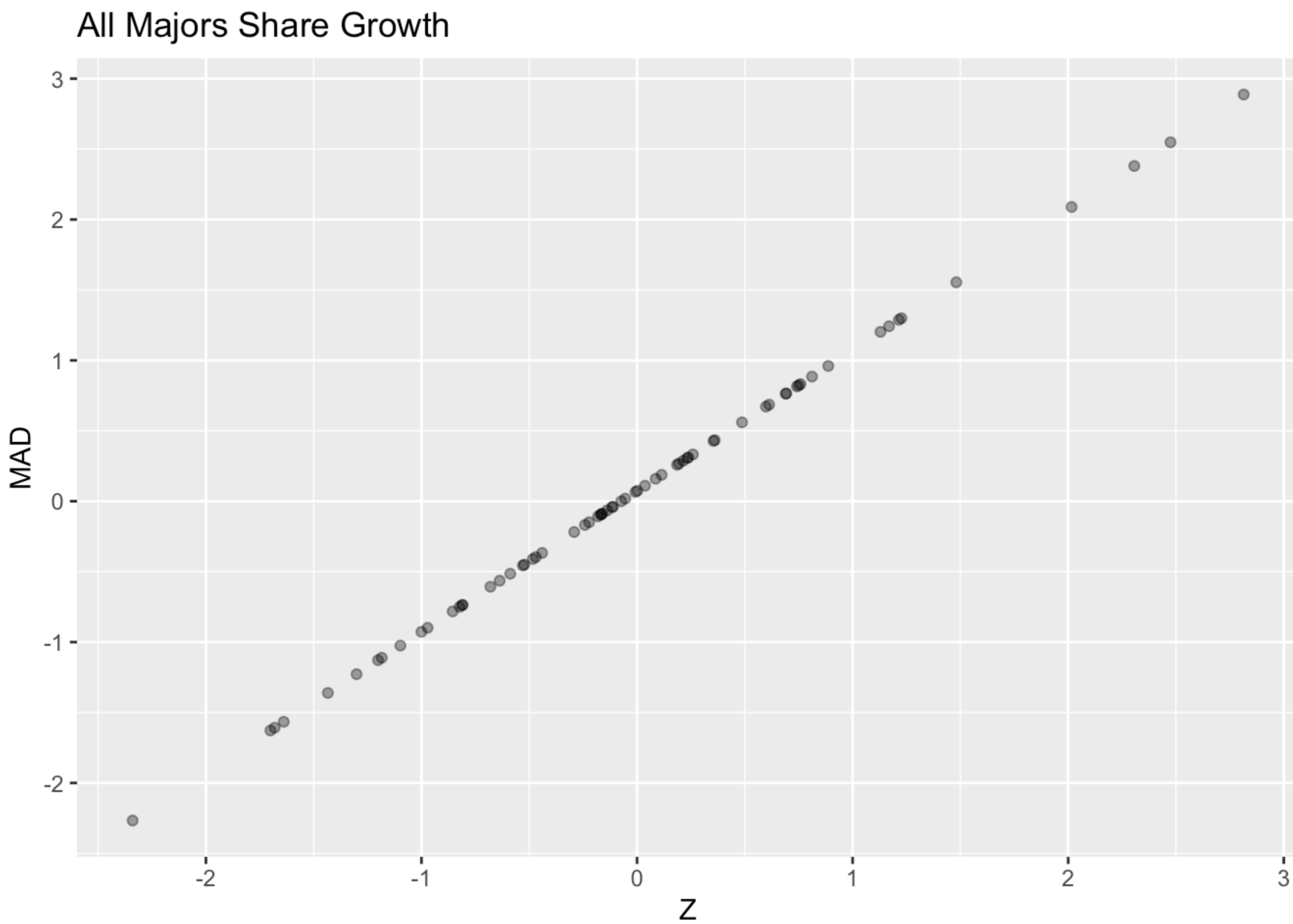
2.2 All Majors Share Growth

2.3 Original Scale



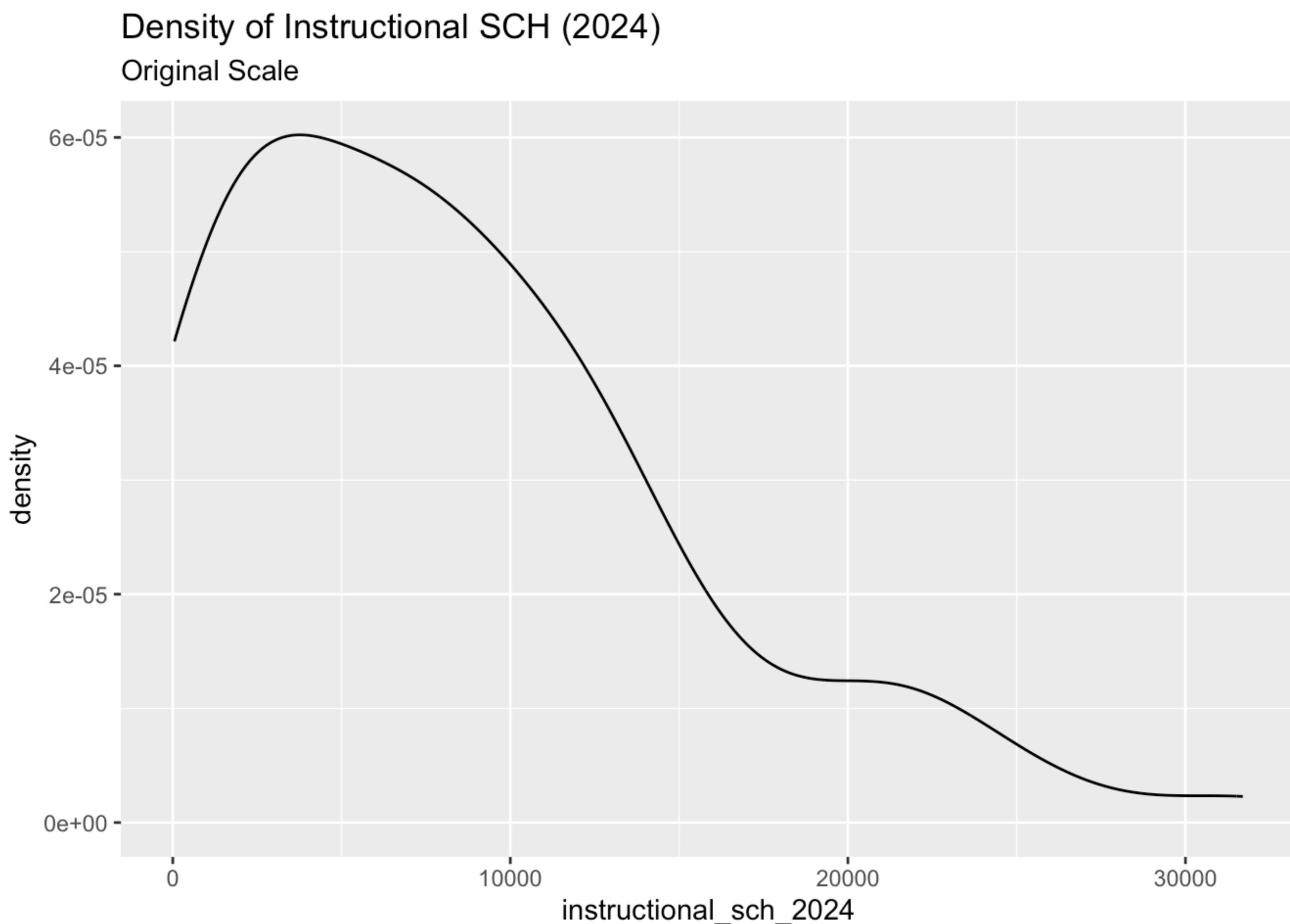
No transformed distribution shown for this metric.

2.3.1 Relationship of Z to Mad

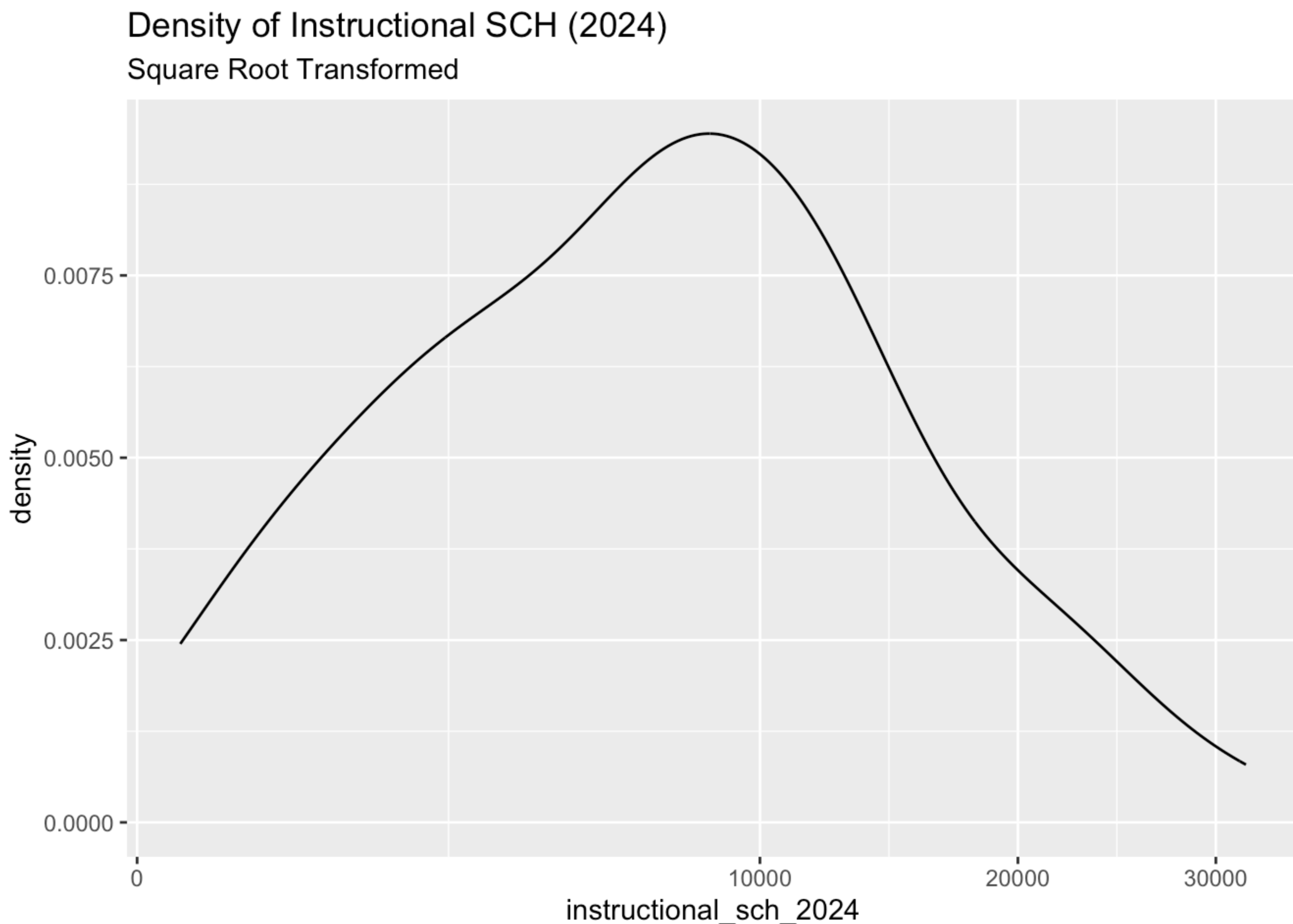


2.4 Instructional SCH (2024)

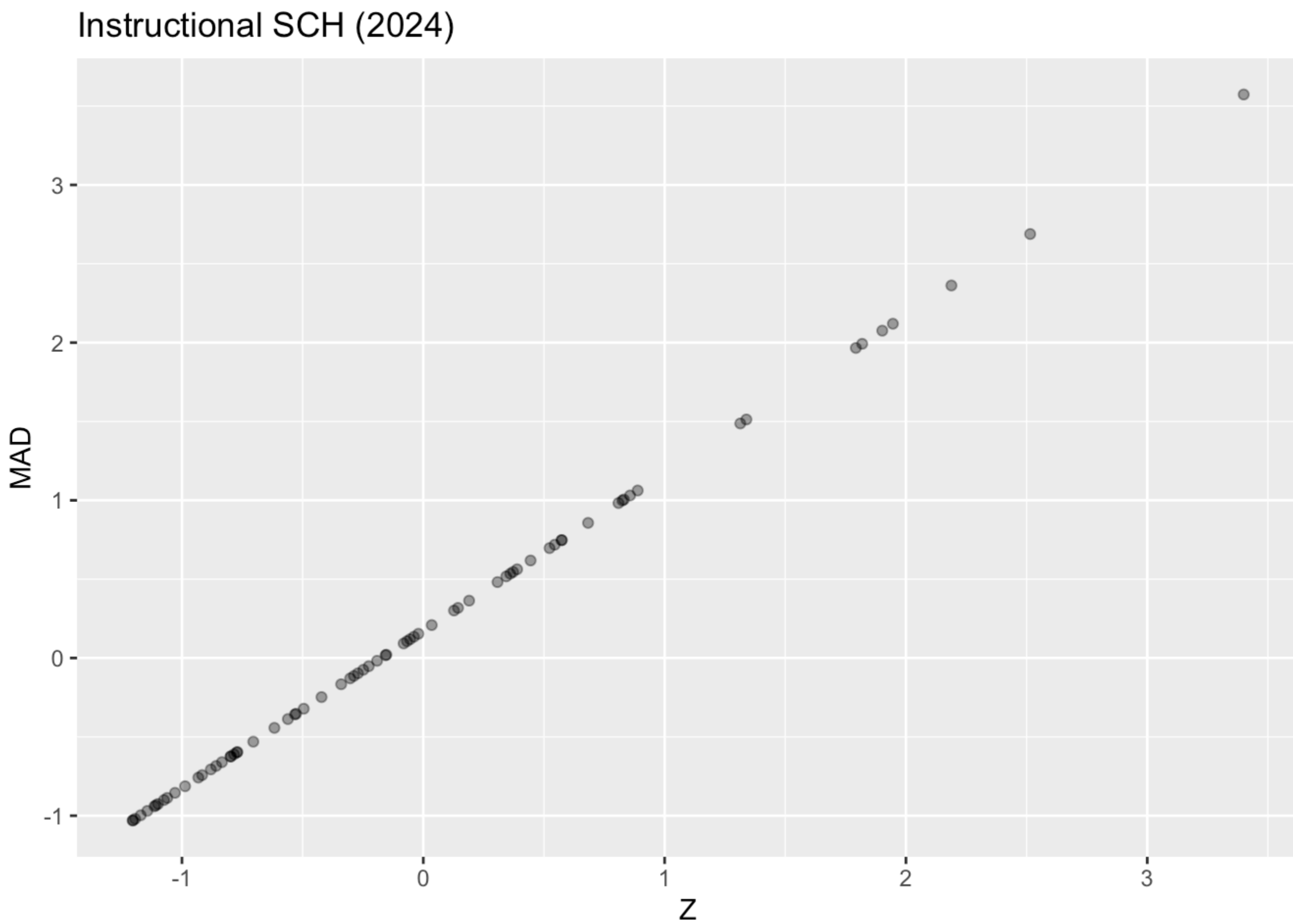
2.4.1 Original Scale



2.4.2 Square-root Transformed

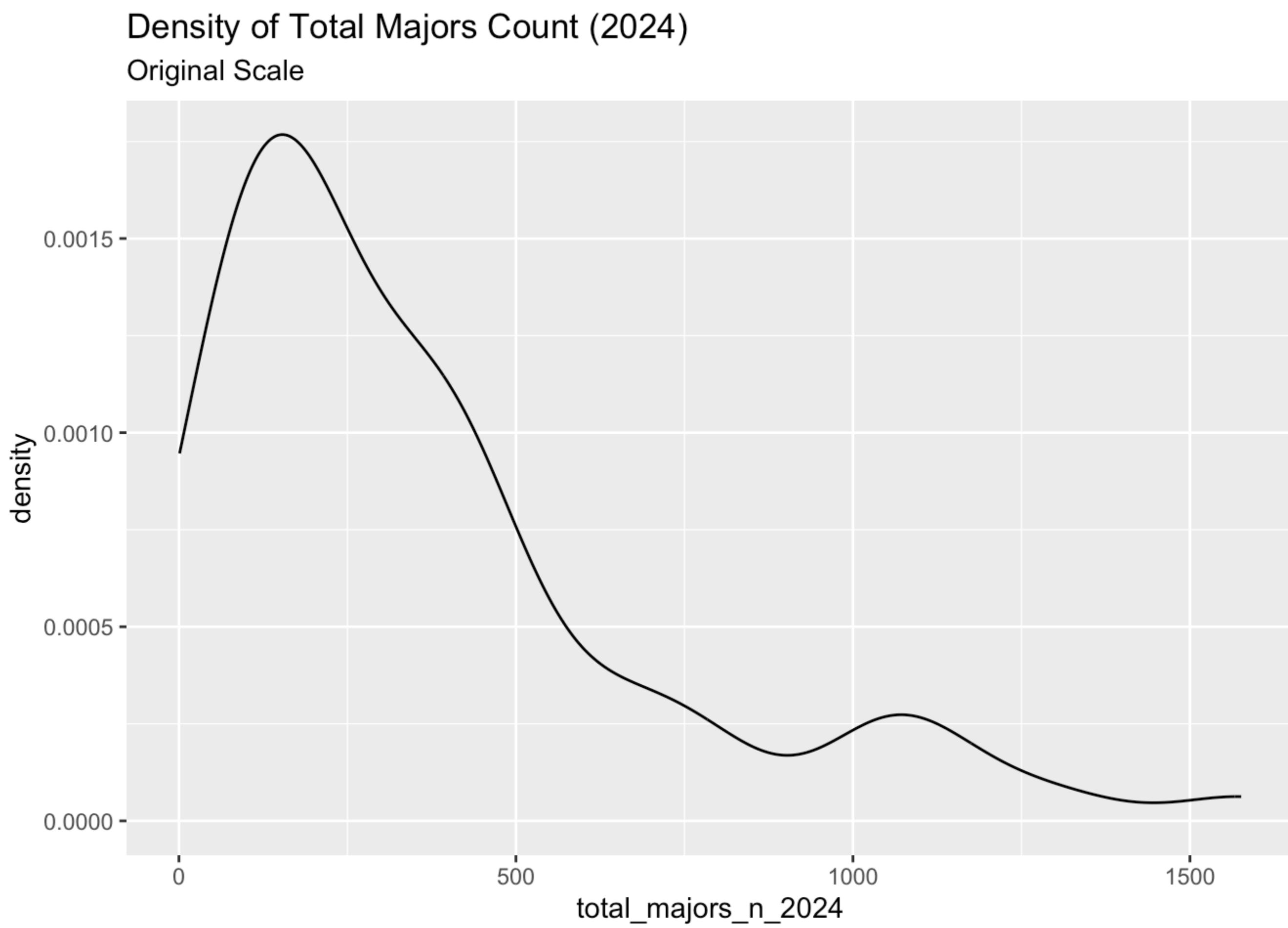


2.4.3 Relationship of Z to Mad

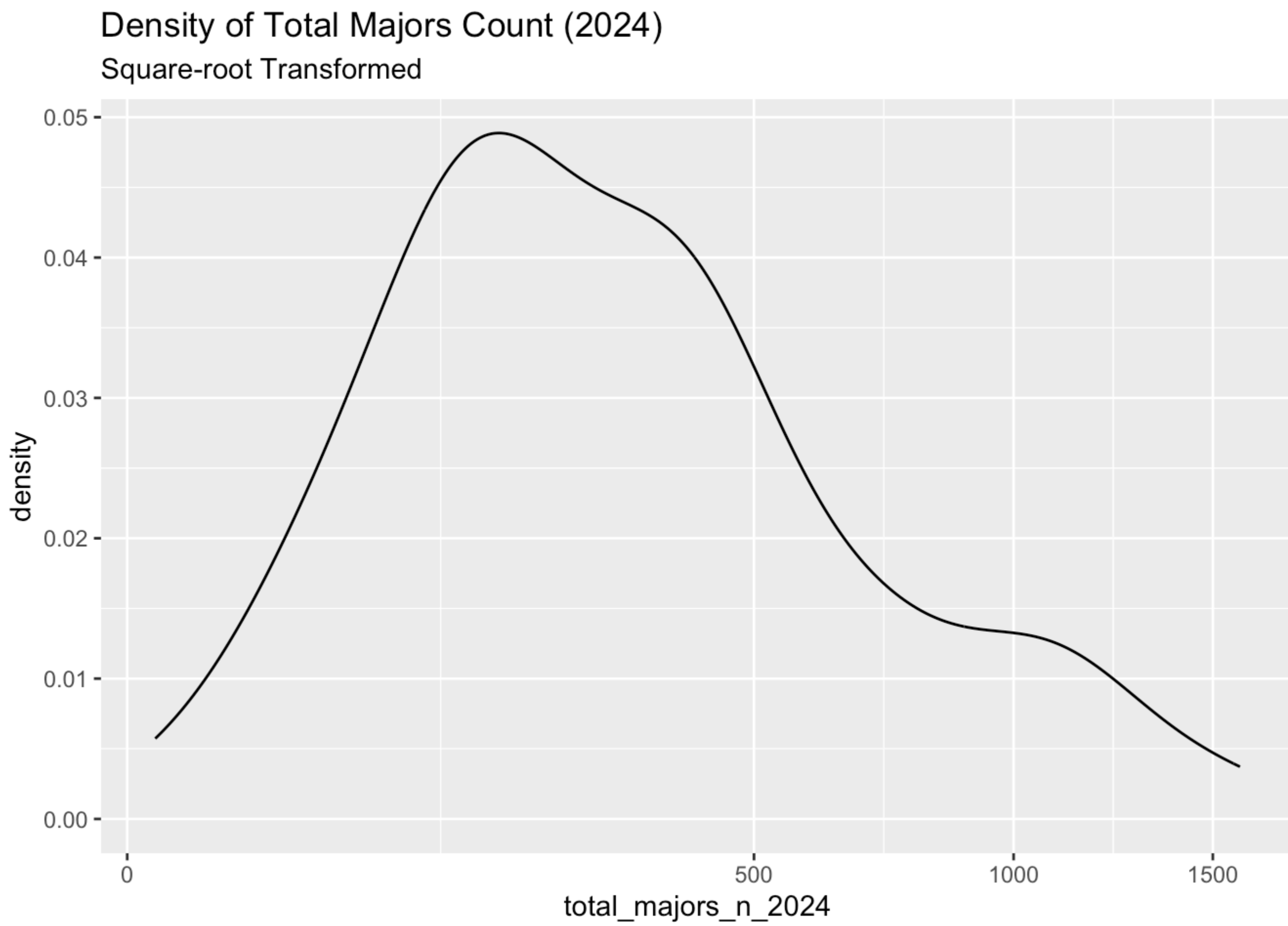


2.5 Total Majors Count (2024)

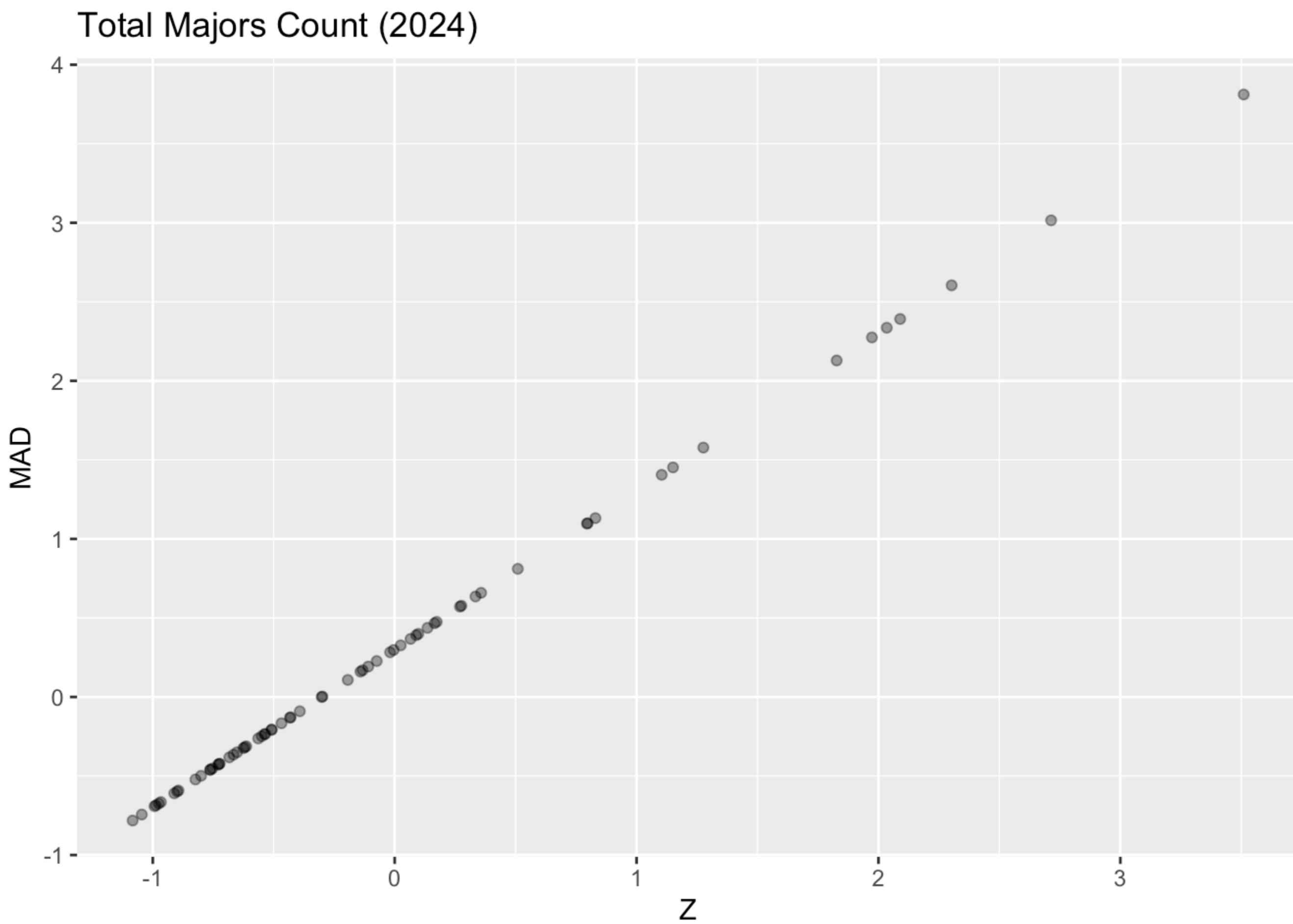
2.5.1 Original Scale



2.5.2 Square-root Transformed

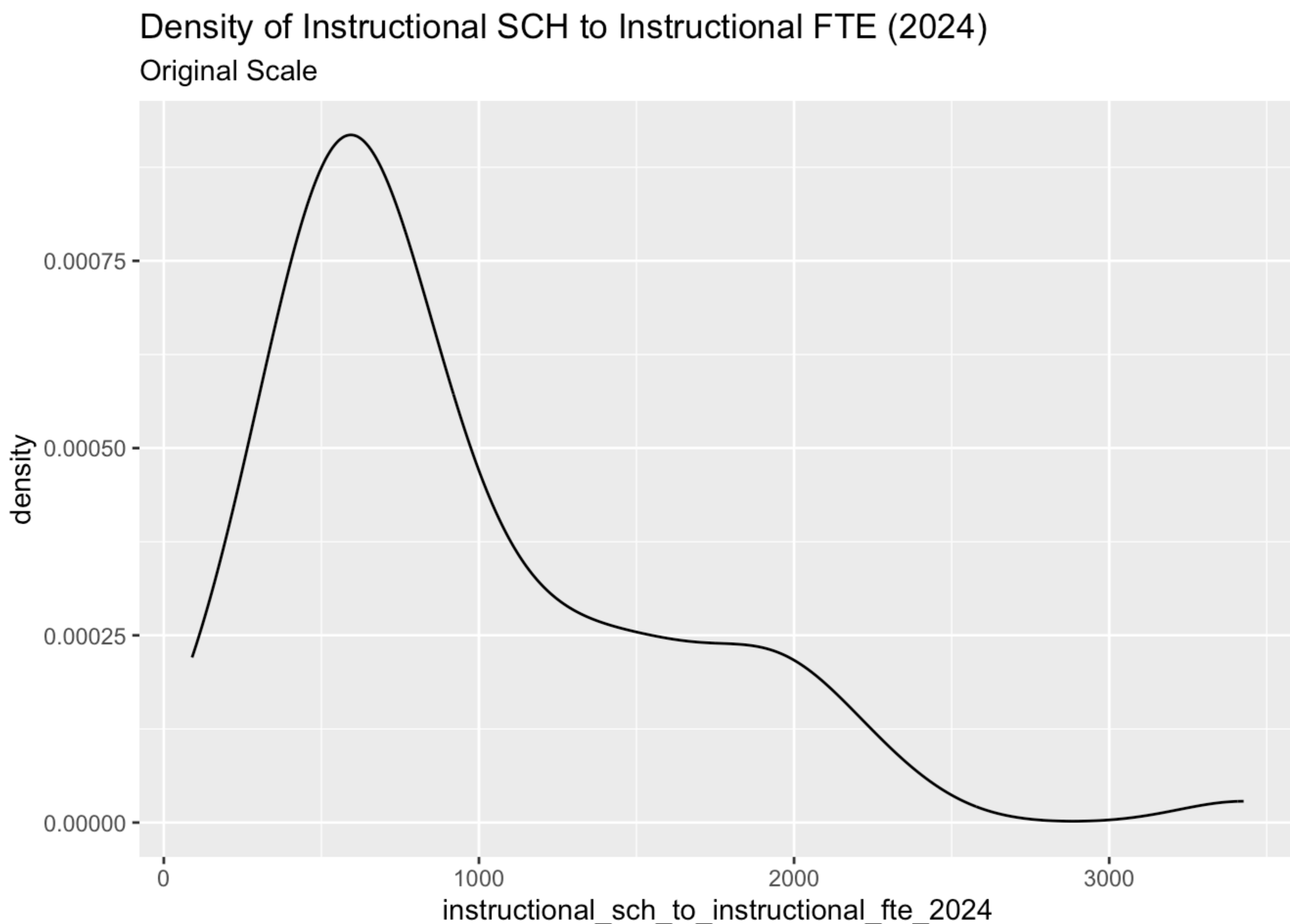


2.5.3 Relationship of Z to Mad

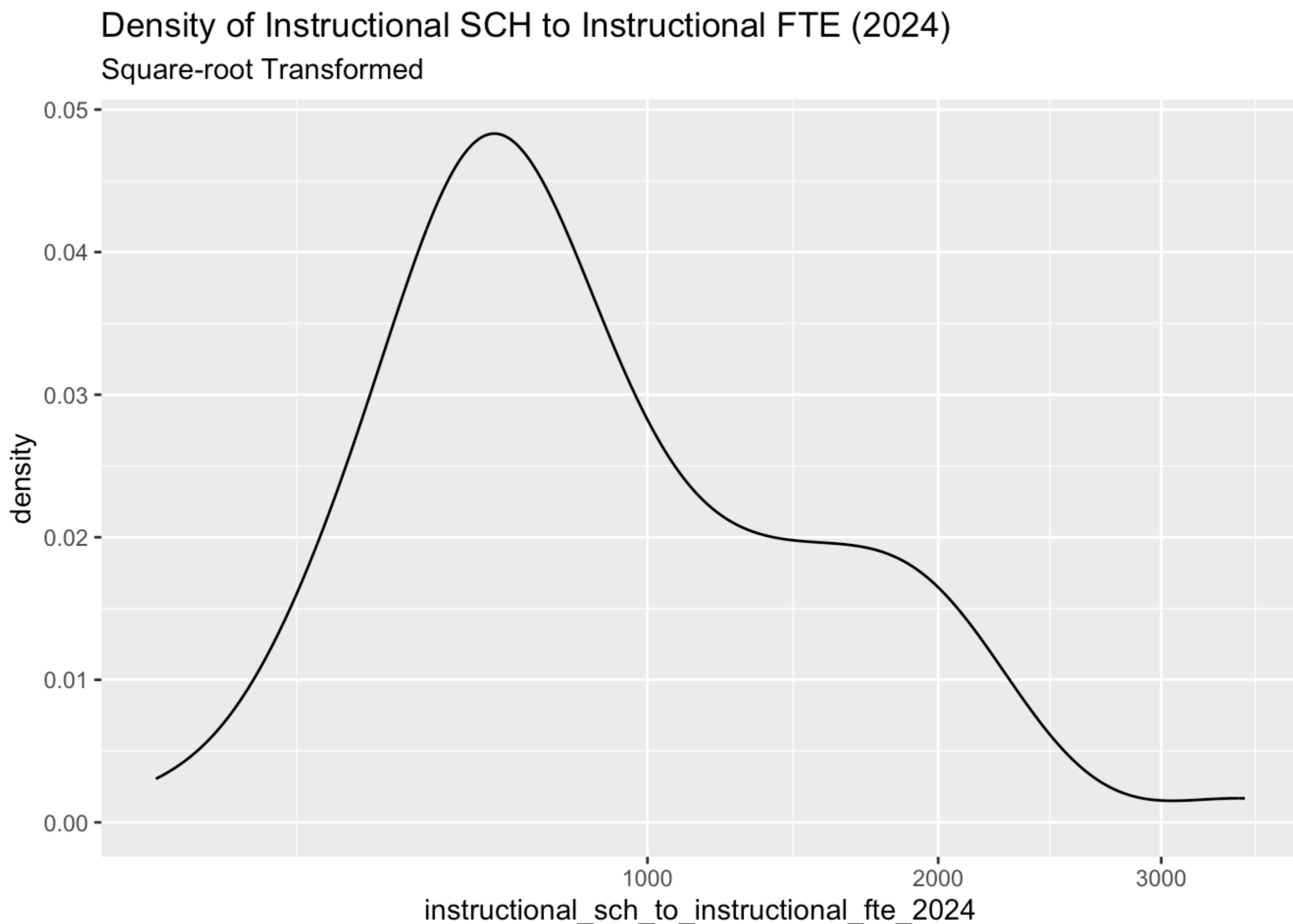


2.6 Instructional SCH to Instructional FTE (2024)

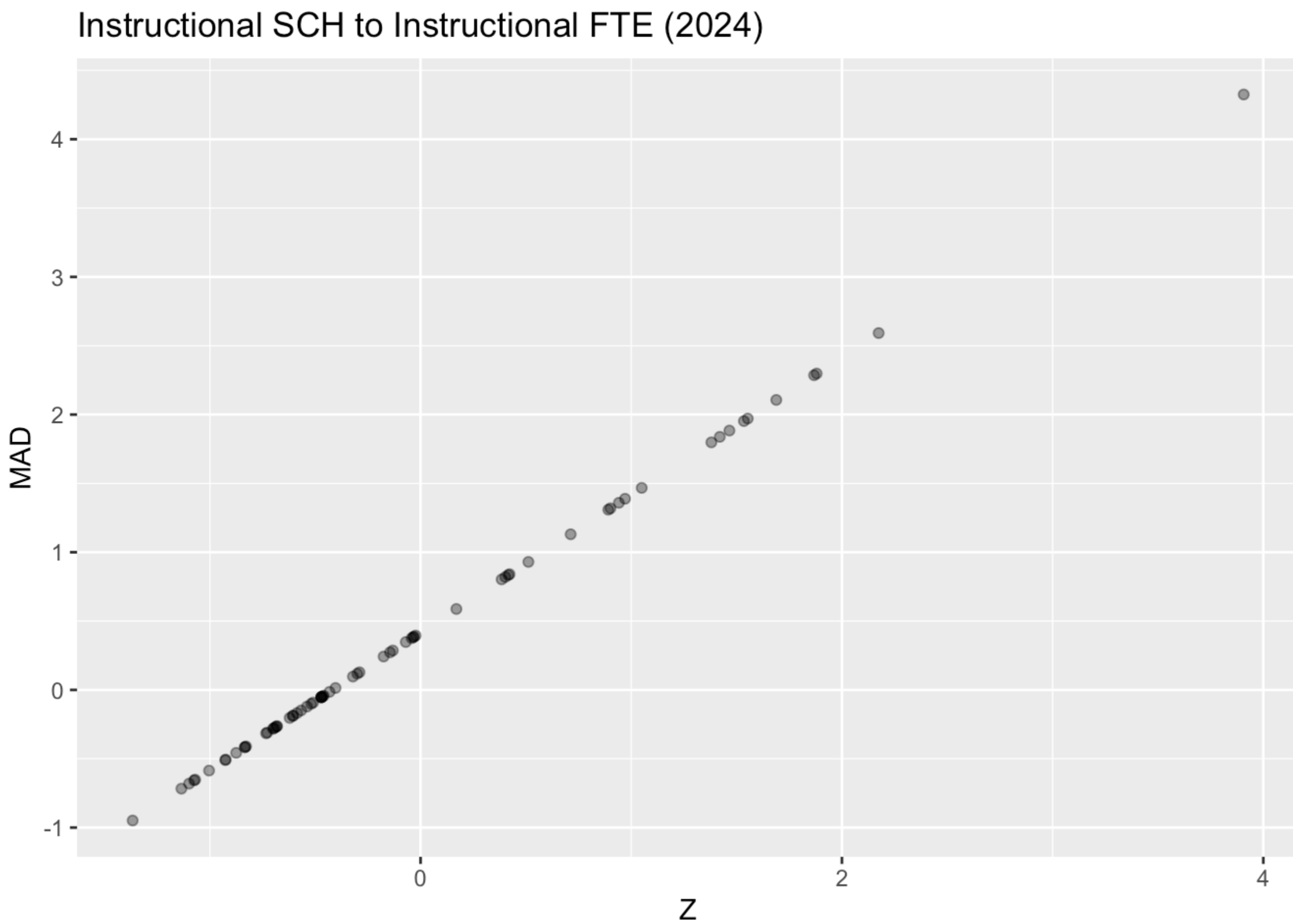
2.6.1 Original Scale



2.6.2 Square-root Transformed

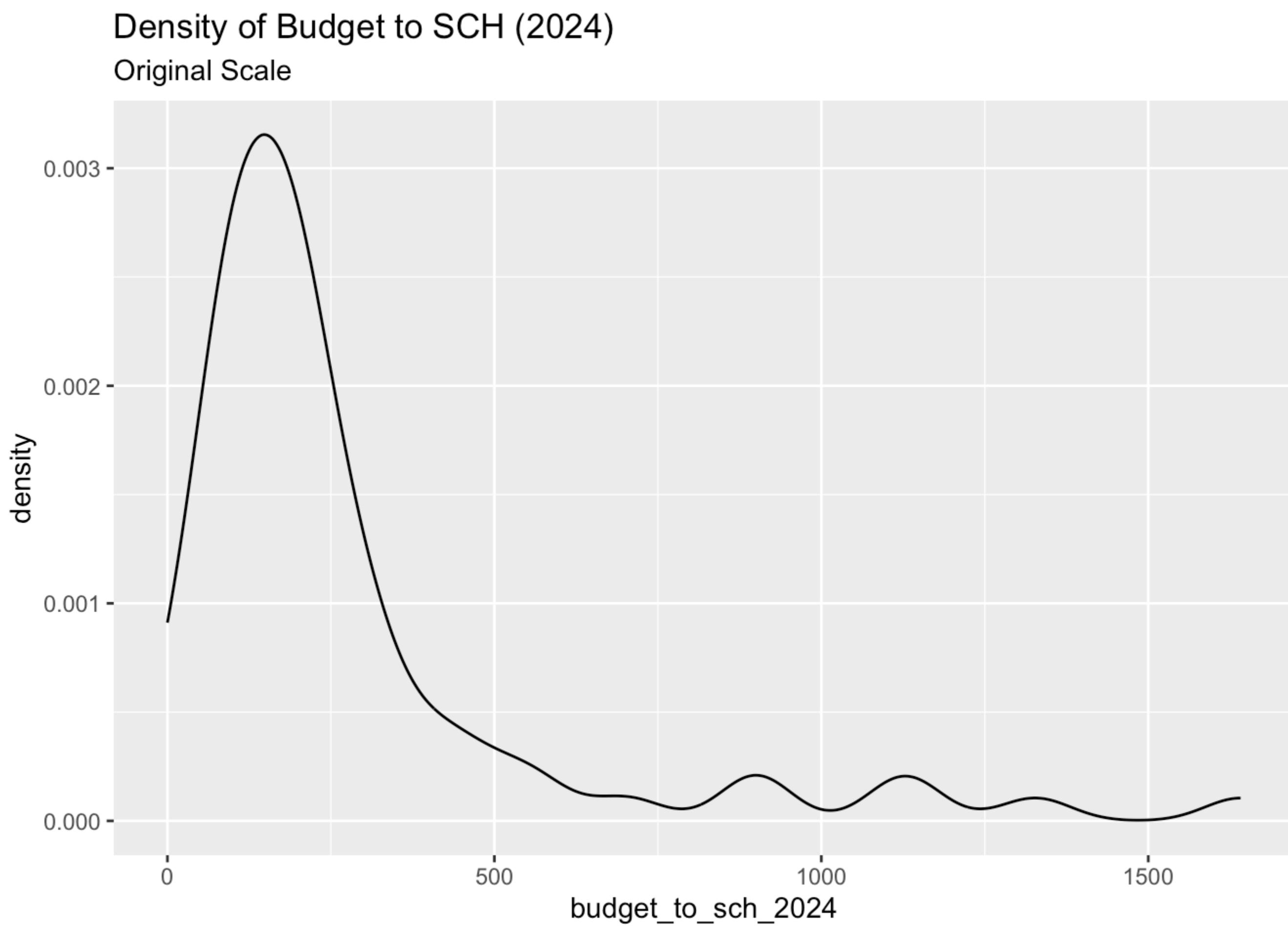


2.6.3 Relationship of Z to Mad

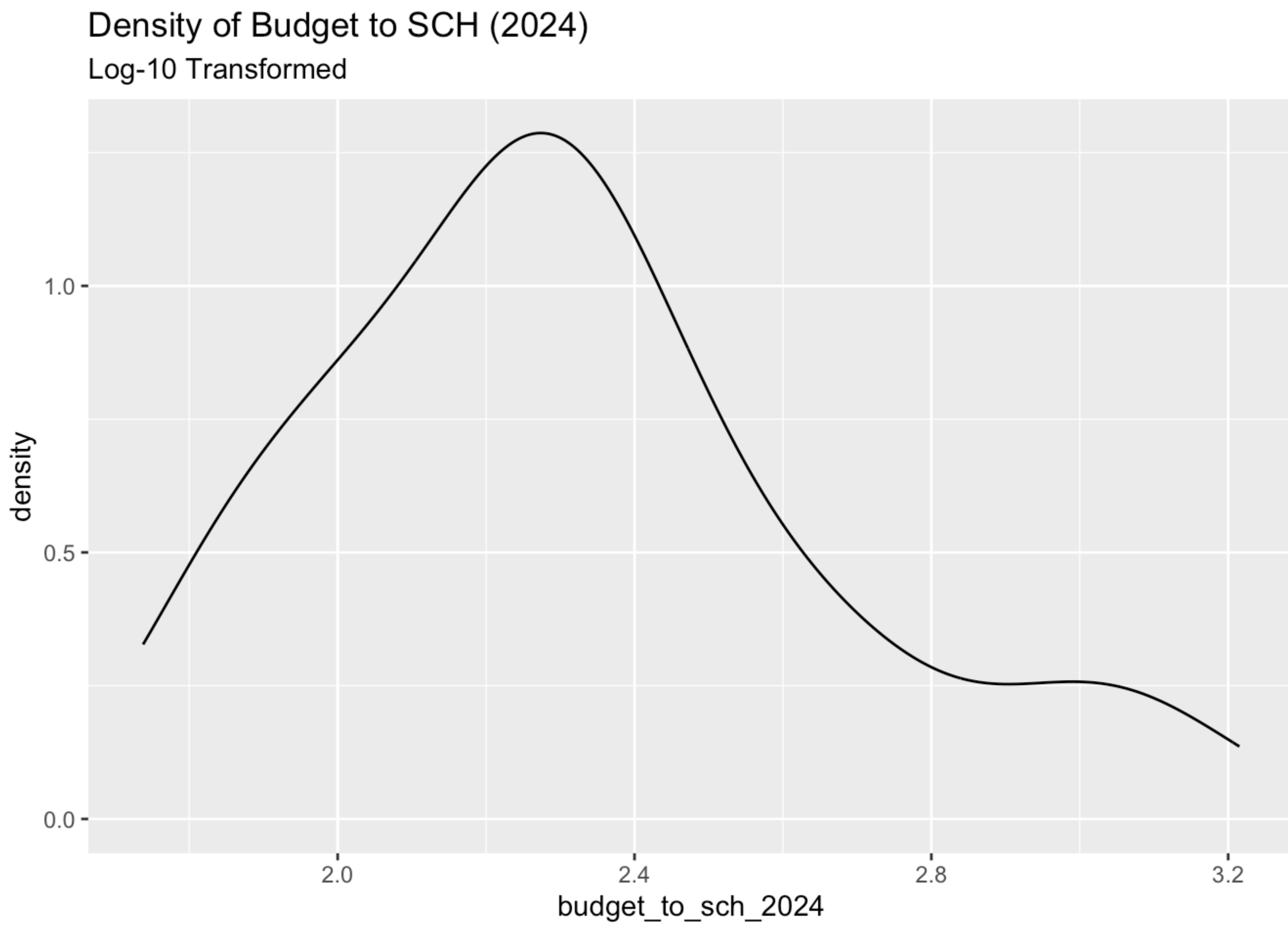


2.7 Budget to SCH (2024)

2.7.1 Original Scale

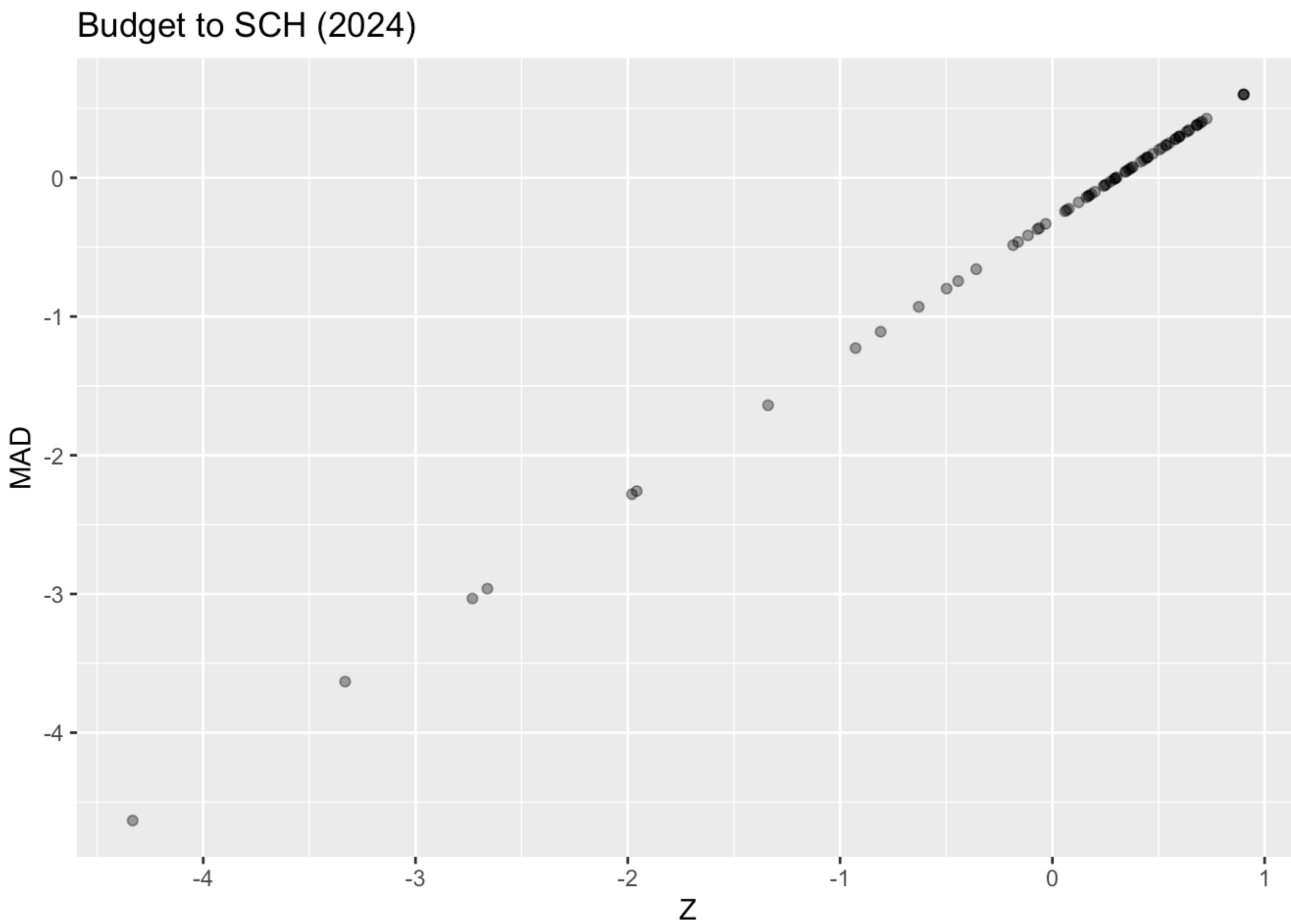


2.7.2 Log-10 Transformed



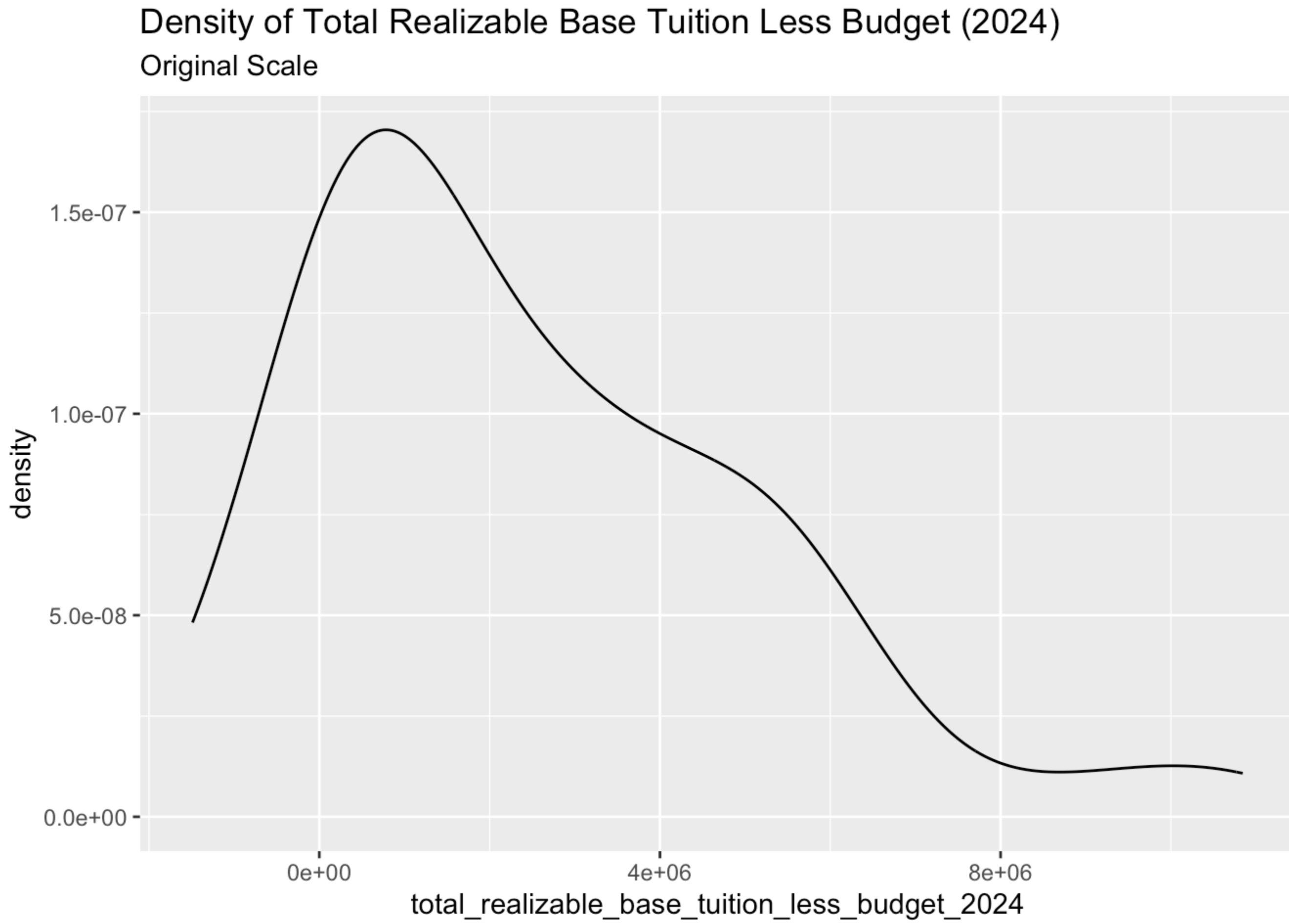
2.7.3 Relationship of Z to Mad

Note that Budget to SCH was reverse-scaled for calculation of z- and MAD-scores.



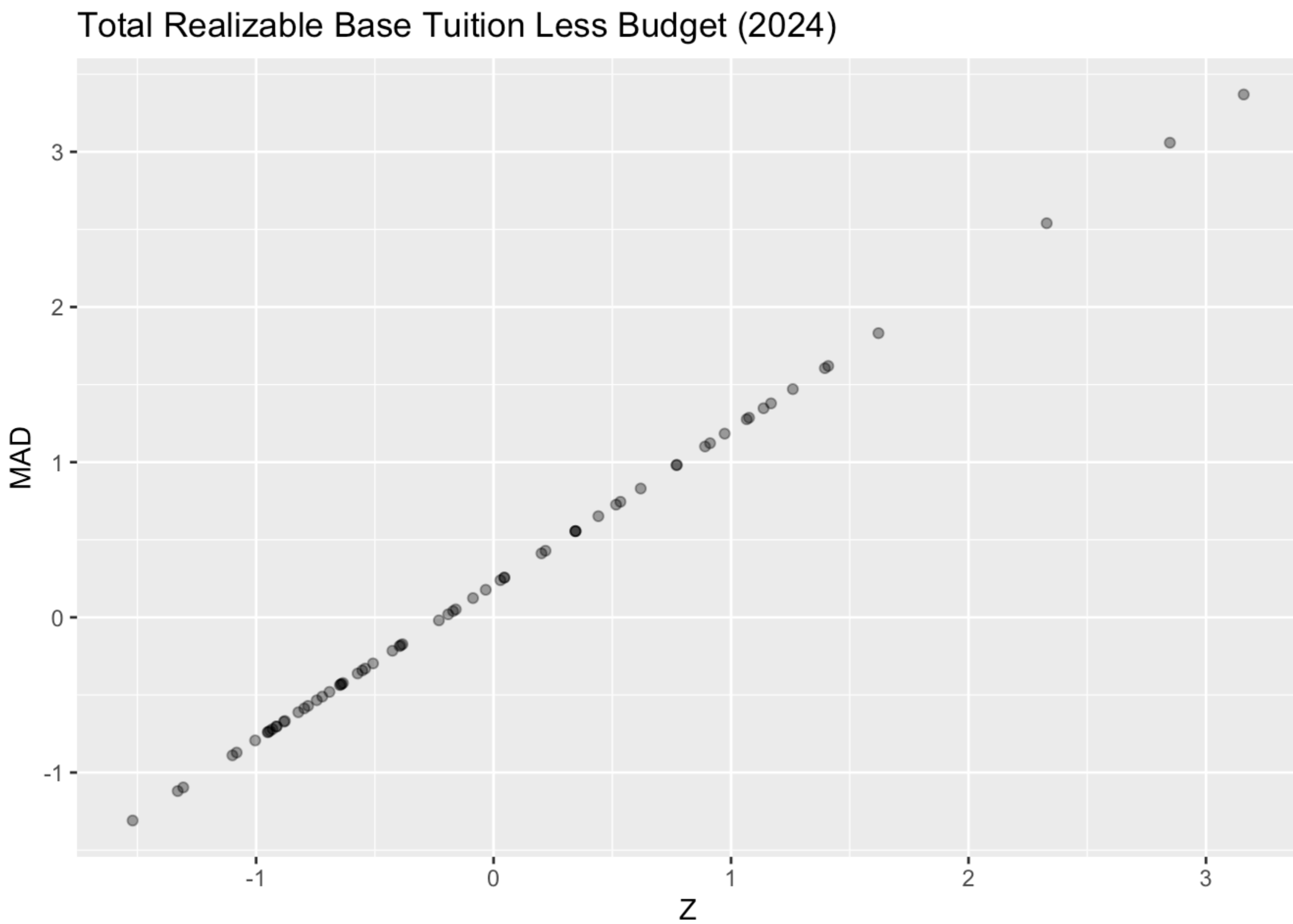
2.8 Total Realizable Base Tuition Less Budget (2024)

2.8.1 Original Scale



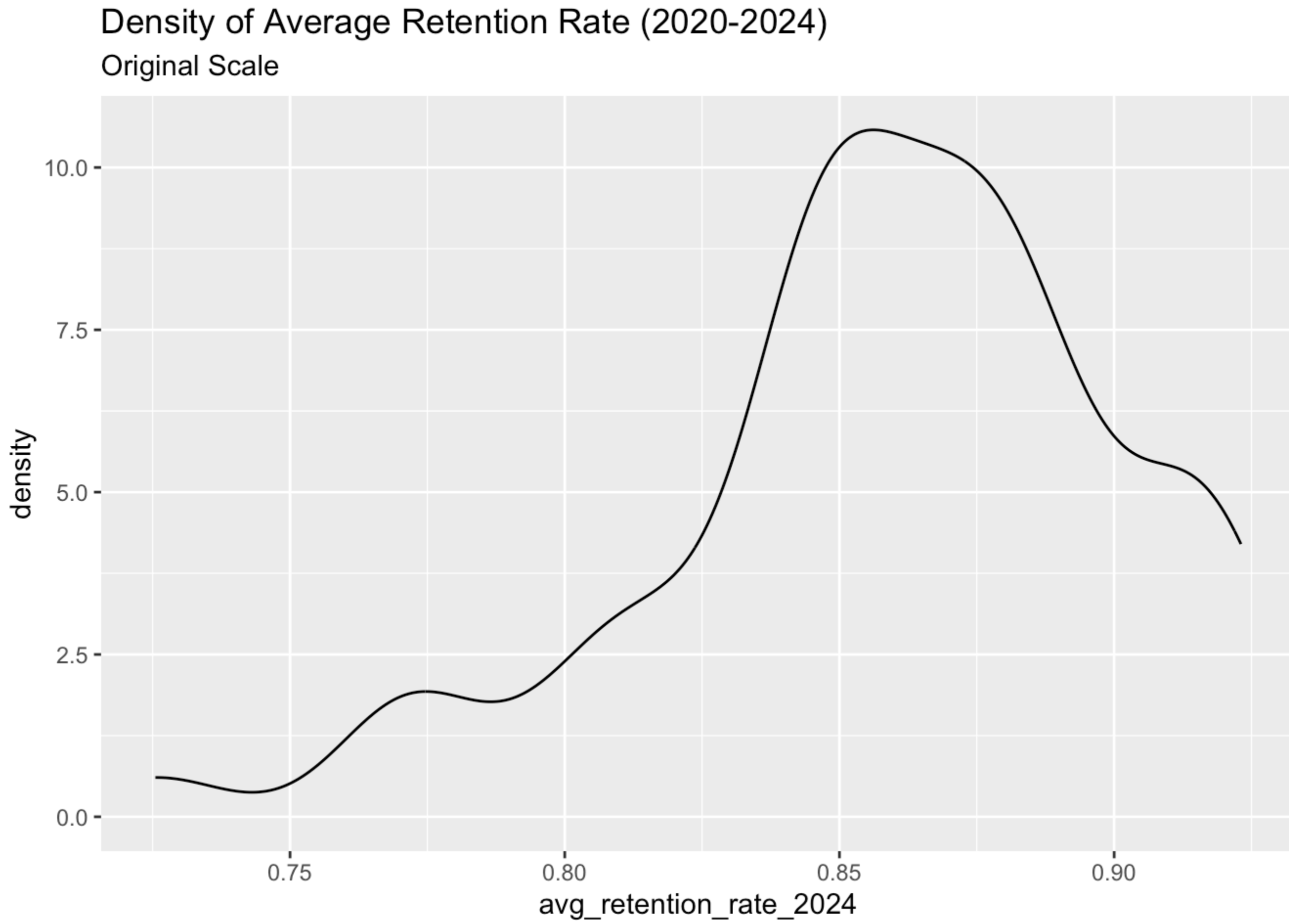
No transformed distribution shown for this metric.

2.8.2 Relationship of Z to Mad

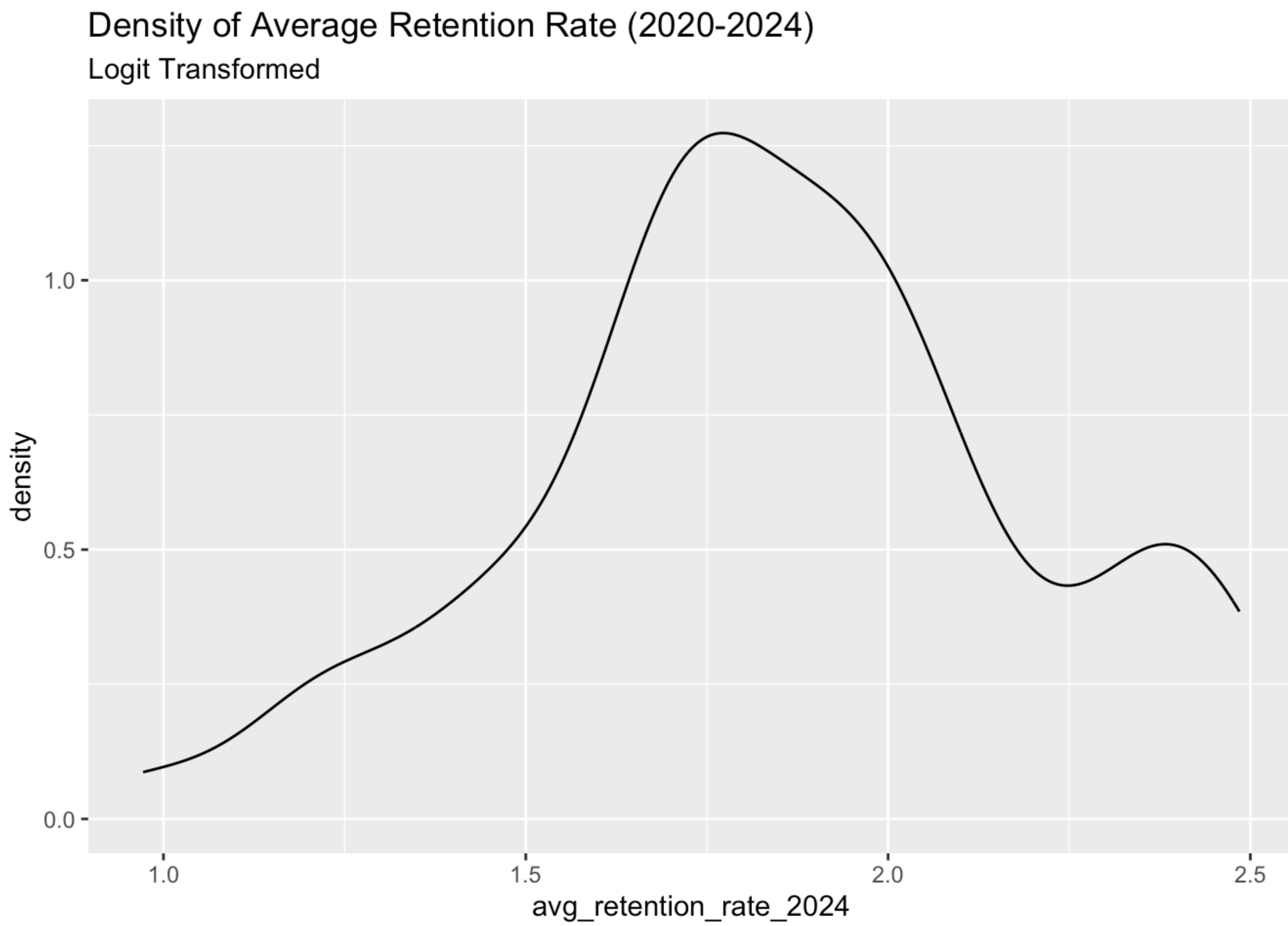


2.9 Average Retention Rate (2020-2024)

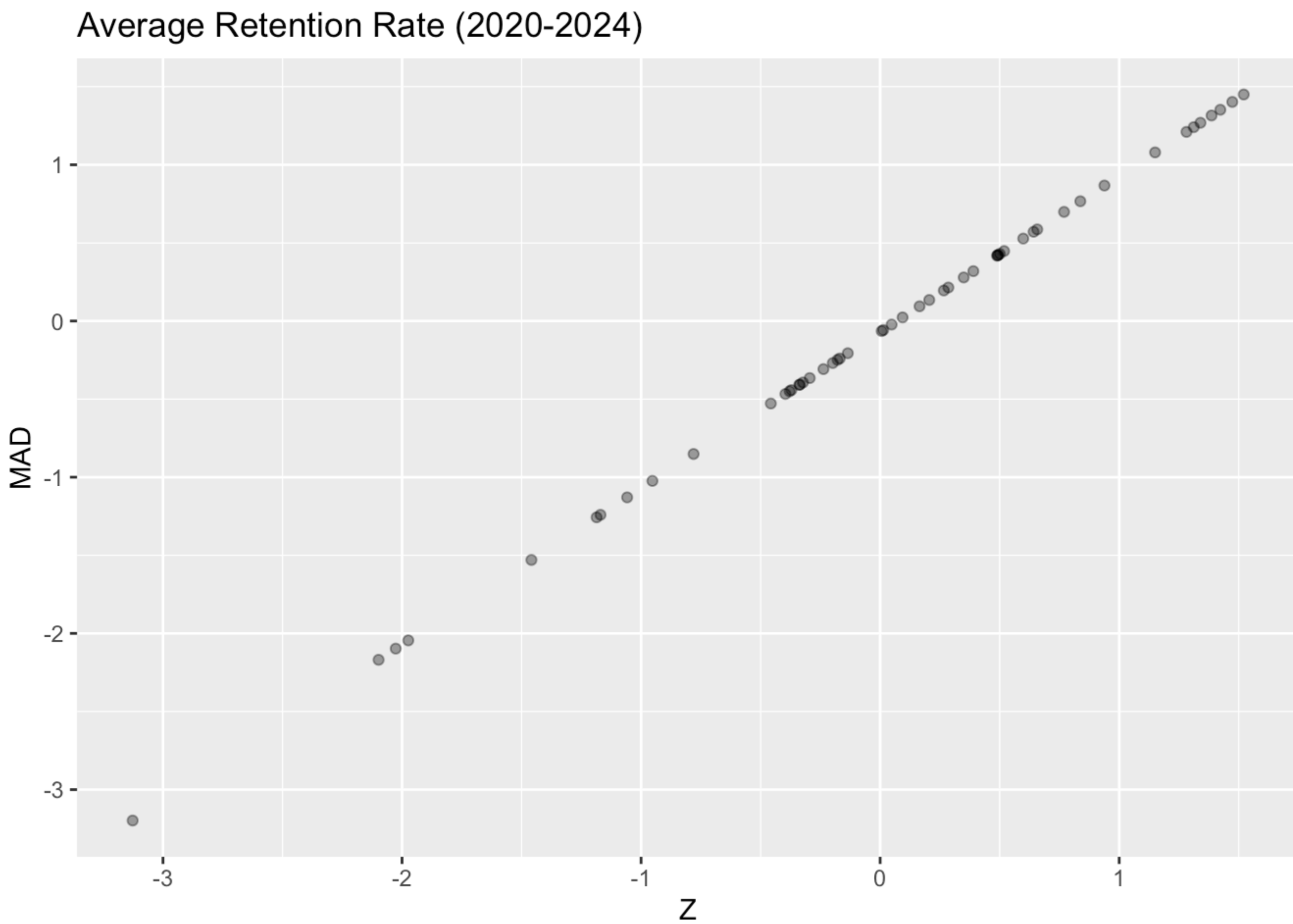
2.9.1 Original Scale



2.9.2 Logit Transformed

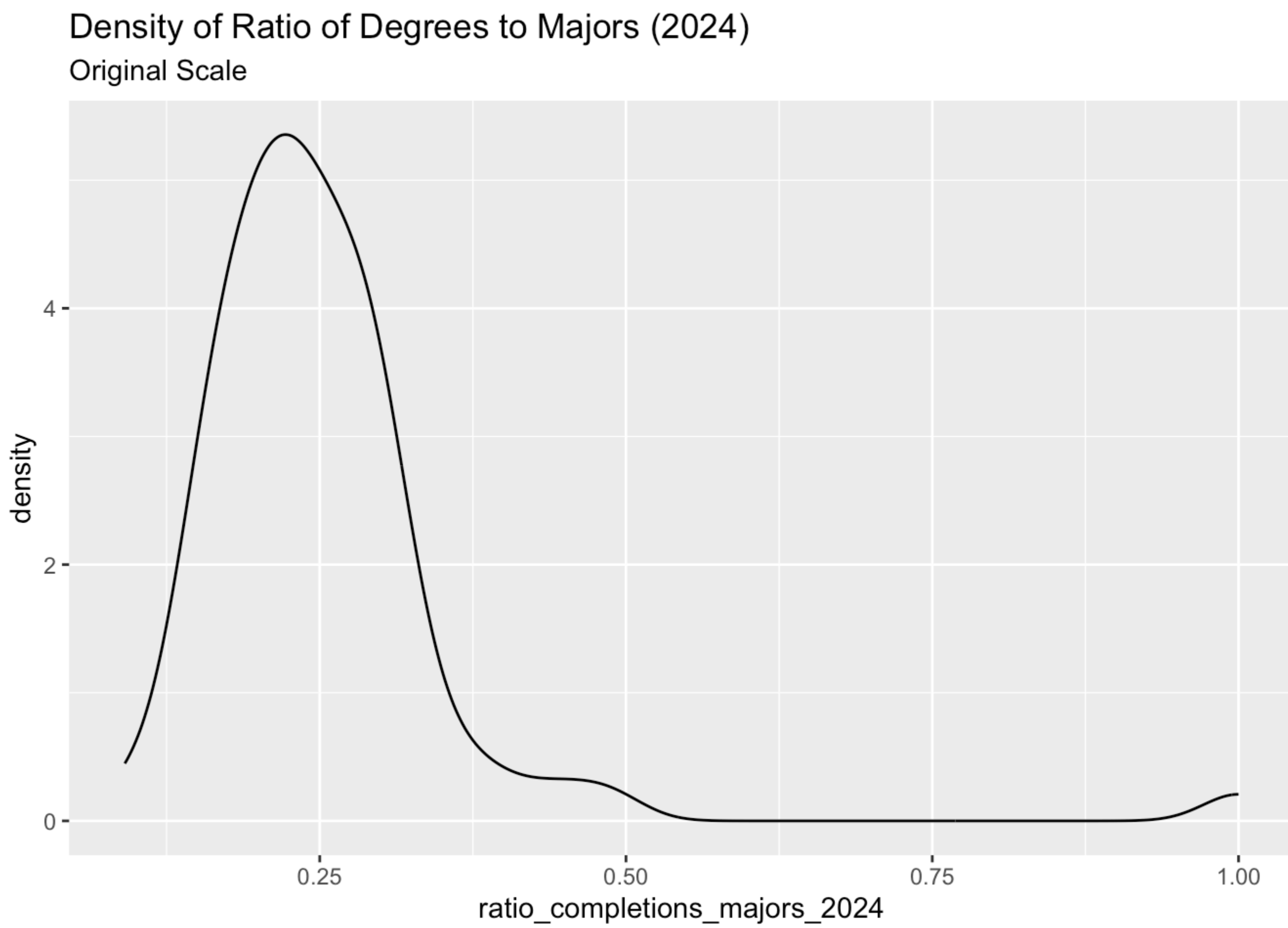


2.9.3 Relationship of Z to Mad

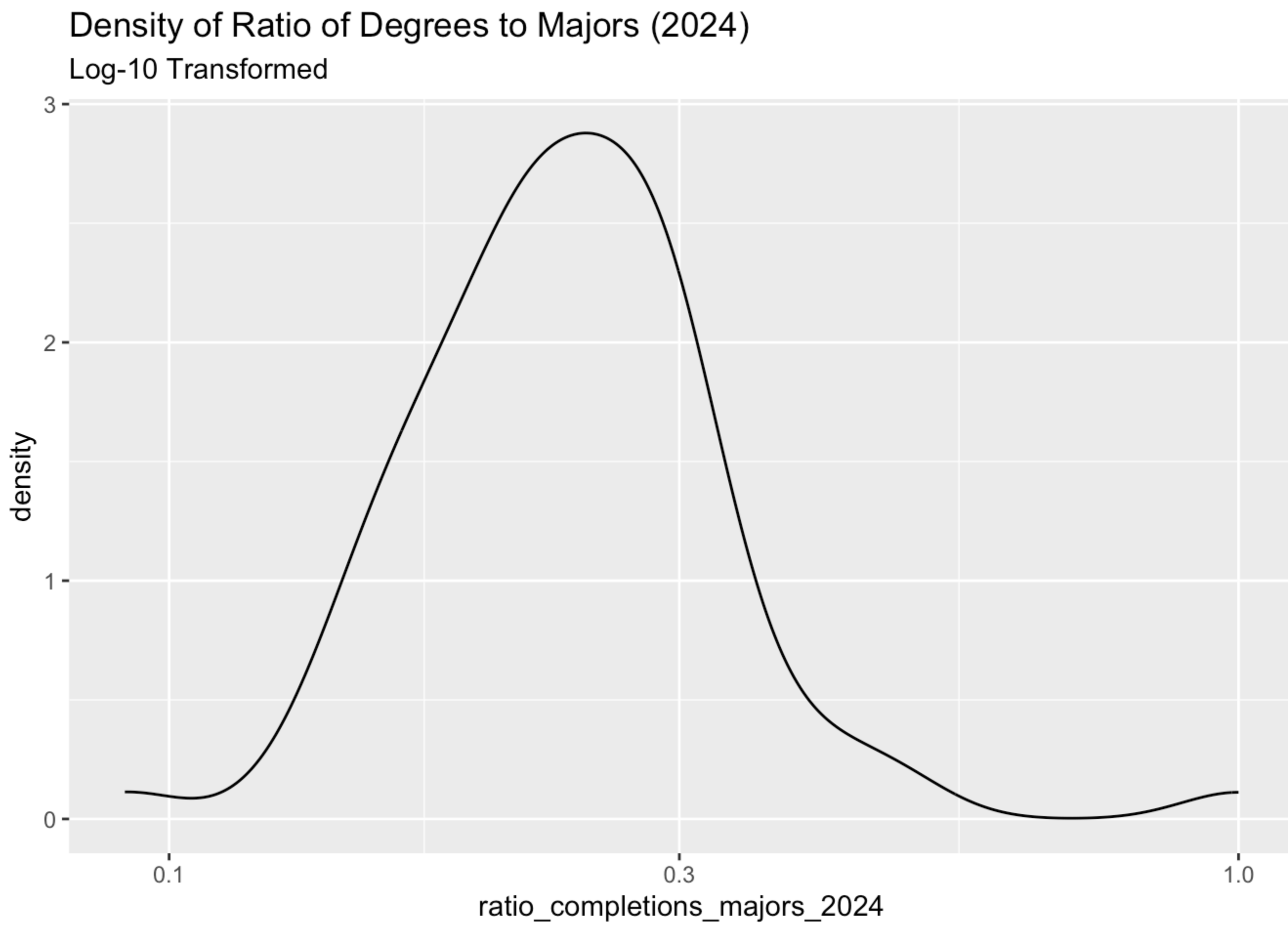


2.10 Ratio of Degrees to Majors (2024)

2.10.1 Original Scale



2.10.2 Log-10 Transformed



2.10.3 Relationship of Z to Mad